

1.- Simplificar los siguientes radicales:

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|--|-------------------------------------|-------------------------------|----------------------------------|--------------------------------|
| 1. $\sqrt{18}$ | 3. $\sqrt[3]{16}$ | 5. $2\sqrt[4]{243}$ | 7. $3\sqrt{81x^3y^4}$ | 9. $\frac{3}{5}\sqrt{125mn^6}$ |
| 2. $3\sqrt{48}$ | 4. $\frac{1}{2}\sqrt[3]{128}$ | 6. $\sqrt{50a^2b}$ | 8. $\frac{1}{2}\sqrt{108a^5b^7}$ | 10. $2a\sqrt{44a^3b^7c^9}$ |
| 11. $2\sqrt[3]{16x^2y^7}$ | 17. $2xy\sqrt[3]{128x^2y^8}$ | 23. $\sqrt{8x^2y^4+16xy^4}$ | | |
| 12. $\frac{2}{3}\sqrt[3]{27m^2n^8}$ | 18. $\frac{1}{3a}\sqrt{27a^3m^7}$ | 24. $\sqrt{2x^2-4xy+2y^2}$ | | |
| 13. $5a\sqrt[3]{160x^7y^9z^{13}}$ | 19. $\frac{3}{5x}\sqrt[3]{375a^8b}$ | 25. $\sqrt{(a-b)(a^2-b^2)}$ | | |
| 14. $\sqrt[4]{80a^4b^5c^{12}}$ | 20. $\frac{1}{3}\sqrt[4]{81a^4b}$ | 26. $\sqrt{2am^2+4amn+2an^2}$ | | |
| 15. $3\sqrt[4]{5x^8y^{14}z^{16}}$ | 21. $\sqrt{9a+18b}$ | 27. $\sqrt{9a^3-36a^2+36a}$ | | |
| 16. $\frac{2}{5}\sqrt[5]{32x^2y^{11}}$ | 22. $\sqrt{3a^3b^2-3a^2b^2}$ | | | |

2.- Simplifica los siguientes radicales:

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|--------------------------|------------------------------------|---|--------------------------------|---|
| 1. $\sqrt{\frac{1}{5}}$ | 4. $3\sqrt{\frac{1}{6}}$ | 7. $\frac{3}{2}\sqrt{\frac{4a^2}{27y^3}}$ | 10. $\sqrt[3]{\frac{2}{3}}$ | 13. $2b^2\sqrt[3]{\frac{125}{4b^5}}$ |
| 2. $\sqrt{\frac{3}{8}}$ | 5. $\frac{1}{2}\sqrt{\frac{2}{3}}$ | 8. $5\sqrt{\frac{9n}{5m^3}}$ | 11. $5\sqrt[3]{\frac{1}{5}}$ | 14. $\frac{2}{3}\sqrt[3]{\frac{27x^2}{16a^2b^4}}$ |
| 3. $2\sqrt{\frac{1}{2}}$ | 6. $\sqrt{\frac{a^2}{8x}}$ | 9. $6\sqrt{\frac{5a^3}{24x^2}}$ | 12. $\sqrt[3]{\frac{8}{9x^2}}$ | 15. $2xy\sqrt[4]{\frac{81a^2}{4x^3y}}$ |

3.- Simplificar

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|-------------------|-------------------------|--------------------------------|-------------------------------------|
| 1. $\sqrt[4]{9}$ | 4. $\sqrt[8]{16}$ | 7. $5\sqrt[6]{49a^2b^4}$ | 10. $\sqrt[12]{64m^6n^{18}}$ |
| 2. $\sqrt[6]{4}$ | 5. $3\sqrt[12]{64}$ | 8. $\sqrt[8]{81x^4y^8}$ | 11. $\sqrt[6]{343a^9x^{12}}$ |
| 3. $\sqrt[9]{27}$ | 6. $\sqrt[4]{25a^2b^2}$ | 9. $\sqrt[10]{32x^{10}y^{15}}$ | 12. $\sqrt[15]{m^{10}n^{15}x^{20}}$ |

4.- Introducir factores en los radicales:

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|-----------------|--------------------------|-------------------------|-----------------------------------|
| 1. $2\sqrt{3}$ | 4. $\frac{1}{2}\sqrt{2}$ | 7. $ab^2\sqrt[3]{a^2b}$ | 10. $(a+b)\sqrt{\frac{a}{a+b}}$ |
| 2. $3\sqrt{5}$ | 5. $3a\sqrt{2a^2}$ | 8. $4m\sqrt[3]{2m^2}$ | 11. $(x+1)\sqrt{\frac{2x}{x+1}}$ |
| 3. $5a\sqrt{b}$ | 6. $5x^2y\sqrt{3}$ | 9. $2a\sqrt[4]{8ab^3}$ | 12. $(x-1)\sqrt{\frac{x-2}{x-1}}$ |

5.- Reducir los siguientes radicales a índice común:

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|--|---|--|
| 1. $\sqrt{5}, \sqrt[3]{2}$ | 5. $\sqrt{5x}, \sqrt[3]{4x^2y}, \sqrt[6]{7a^3b}$ | 9. $\sqrt[4]{3a}, \sqrt[5]{2b^2}, \sqrt[10]{7x^3}$ |
| 2. $\sqrt{2}, \sqrt[4]{3}$ | 6. $\sqrt[3]{2ab}, \sqrt[5]{3a^2x}, \sqrt[15]{5a^3x^2}$ | 10. $2\sqrt[3]{a}, 3\sqrt{2b}, 4\sqrt[4]{5x^2}$ |
| 3. $\sqrt{3}, \sqrt[3]{4}, \sqrt[4]{8}$ | 7. $\sqrt[4]{8a^2x^3}, \sqrt[6]{3a^5m^4}$ | 11. $3\sqrt[3]{a^2}, \frac{1}{2}\sqrt[6]{b^3}, 4\sqrt[9]{x^5}$ |
| 4. $\sqrt{2}, \sqrt[3]{3}, \sqrt[4]{5}, \sqrt[6]{7}$ | 8. $\sqrt[3]{x^2}, \sqrt[6]{2y^3}, \sqrt[9]{5m^7}$ | 12. $\sqrt{2m}, 3\sqrt[5]{a^3x^4}, 2\sqrt[10]{x^7y^2}$ |

6.- Ordena de mayor a menor los siguientes números:

1. $\sqrt{5}, \sqrt[3]{2}$

3. $\sqrt{11}, \sqrt[3]{43}$

5. $\sqrt[4]{3}, \sqrt[5]{4}, \sqrt[10]{15}$

2. $\sqrt[6]{15}, \sqrt[4]{7}$

4. $\sqrt{3}, \sqrt[3]{5}, \sqrt[6]{32}$

6. $\sqrt[3]{2}, \sqrt[6]{3}, \sqrt[9]{9}$

7.- Opera los siguientes radicales semejantes:

1. $7\sqrt{2} - 15\sqrt{2}$

5. $\frac{3}{4}\sqrt{2} - \frac{1}{2}\sqrt{2}$

8. $\frac{1}{4}\sqrt{3} + 5\sqrt{3} - \frac{1}{8}\sqrt{3}$

2. $4\sqrt{3} - 20\sqrt{3} + 19\sqrt{3}$

9. $a\sqrt{b} - 3a\sqrt{b} + 7a\sqrt{b}$

3. $\sqrt{5} - 22\sqrt{5} - 8\sqrt{5}$

6. $\frac{3}{5}\sqrt{3} - \sqrt{3}$

10. $3x\sqrt{y} + (a-x)\sqrt{y} - 2x\sqrt{y}$

4. $\sqrt{2} - 9\sqrt{2} + 30\sqrt{2} - 40\sqrt{2}$

7. $2\sqrt{5} - \frac{1}{2}\sqrt{5} + \frac{3}{4}\sqrt{5}$

11. $(x-1)\sqrt{3} + (x-3)\sqrt{3} + 4\sqrt{3}$

12. $\frac{1}{3}\sqrt[3]{2} - \frac{2}{3}\sqrt[3]{2} + 2\sqrt[3]{2}$

13. $\frac{3}{5}\sqrt[3]{2} - \frac{1}{4}\sqrt[3]{2} + \frac{1}{6}\sqrt[3]{2}$

14. $x\sqrt[3]{a^2} - (a-2x)\sqrt[3]{a^2} + (2a-3x)\sqrt[3]{a^2}$

8.- Opera

1. $\sqrt{45} - \sqrt{27} - \sqrt{20}$

11. $5\sqrt{128} - \frac{1}{3}\sqrt{\frac{1}{3}} - 5\sqrt{98} + \sqrt{\frac{1}{27}}$

2. $\sqrt{175} + \sqrt{243} - \sqrt{63} - 2\sqrt{75}$

12. $2\sqrt{700} - 15\sqrt{\frac{1}{45}} + 4\sqrt{\frac{5}{16}} - 56\sqrt{\frac{1}{7}}$

3. $\sqrt{80} - 2\sqrt{252} + 3\sqrt{405} - 3\sqrt{500}$

13. $\sqrt{25ax^2} + \sqrt{49b} - \sqrt{9ax^2}$

4. $7\sqrt{450} - 4\sqrt{320} + 3\sqrt{80} - 5\sqrt{800}$

14. $2\sqrt{m^2n} - \sqrt{9m^2n} + \sqrt{16mn^2} - \sqrt{4m}$

5. $\frac{1}{2}\sqrt{12} - \frac{1}{3}\sqrt{18} + \frac{3}{4}\sqrt{48} + \frac{1}{6}\sqrt{72}$

15. $a\sqrt{320x} - 7\sqrt{5a^2x} - (a-4b)\sqrt{5x}$

6. $\frac{3}{4}\sqrt{176} - \frac{2}{3}\sqrt{45} + \frac{1}{8}\sqrt{320} + \frac{1}{5}\sqrt{275}$

16. $\sqrt{9x-9} + \sqrt{4x-4} - 5\sqrt{x-1}$

7. $\frac{1}{7}\sqrt{147} - \frac{1}{5}\sqrt{700} + \frac{1}{10}\sqrt{28} + \frac{1}{3}\sqrt{2187}$

17. $2\sqrt{a^4x+3a^4y} - a^2\sqrt{9x+27y} + \sqrt{25a^4x+75a^4y}$

8. $\sqrt{\frac{1}{3}} - \sqrt{\frac{1}{2}} + \sqrt{\frac{3}{4}}$

18. $3a\sqrt{\frac{a+1}{a^2}} - \sqrt{4a+4} + (a+1)\sqrt{\frac{1}{a+1}}$

9. $\sqrt{\frac{9}{5}} - \sqrt{\frac{1}{6}} - \sqrt{\frac{1}{20}} + \sqrt{6}$

19. $(a-b)\sqrt{\frac{a+b}{a-b}} - (a+b)\sqrt{\frac{a-b}{a+b}} + (2a-2b)\sqrt{\frac{1}{a-b}}$

10. $\frac{5}{3}\sqrt{\frac{3}{5}} - \frac{1}{2}\sqrt{\frac{3}{4}} - 5\sqrt{\frac{1}{15}} + 3\sqrt{\frac{1}{12}}$

9.- Desarrolla:

1. $(4\sqrt{2})^2$

3. $(5\sqrt{7})^2$

5. $(3\sqrt[3]{2a^2b})^4$

7. $(\sqrt[5]{81ab^3})^3$

9. $(4a\sqrt{2x})^2$

11. $(3\sqrt{x-a})^2$

2. $(2\sqrt{3})^2$

4. $(2\sqrt[3]{4})^2$

6. $(\sqrt[4]{8x^3})^2$

8. $(\sqrt[6]{18})^3$

10. $(2\sqrt{x+1})^2$

12. $(4\sqrt[6]{9a^3b^4})^3$

Elevar al cuadrado

13. $\sqrt{2} - \sqrt{3}$

15. $\sqrt{5} - \sqrt{7}$

17. $\sqrt{x} + \sqrt{x-1}$

19. $\sqrt{a+1} - \sqrt{a-1}$

14. $4\sqrt{2} + \sqrt{3}$

16. $5\sqrt{7} - 6$

18. $\sqrt{x+1} - 4\sqrt{x}$

20. $2\sqrt{2x-1} + \sqrt{2x+1}$

10.- Calcula:

1. $\sqrt[3]{54} - \sqrt[3]{24} - \sqrt[3]{16}$

2. $\sqrt[3]{40} + \sqrt[3]{1029} - \sqrt[3]{625}$

3. $2\sqrt[3]{250} - 4\sqrt[3]{24} - 6\sqrt[3]{16} + \sqrt[3]{2187}$

4. $5\sqrt[3]{48} - 3\sqrt[3]{3645} - 2\sqrt[3]{384} + 4\sqrt[3]{1715}$

5. $\sqrt[3]{81} - 3\sqrt[3]{375} + \sqrt[3]{686} + 2\sqrt[3]{648}$

6. $\frac{1}{2}\sqrt[3]{24} - \frac{2}{3}\sqrt[3]{54} + \frac{3}{5}\sqrt[3]{375} - \frac{1}{4}\sqrt[3]{128}$

7. $\frac{3}{5}\sqrt[3]{625} - \frac{3}{2}\sqrt[3]{192} + \frac{1}{7}\sqrt[3]{1715} - \frac{3}{8}\sqrt[3]{1536}$

8. $\sqrt[3]{\frac{1}{4}} + \sqrt[3]{\frac{1}{3}} - \sqrt[3]{\frac{2}{27}}$

9. $6\sqrt[3]{\frac{1}{24}} + \sqrt[3]{\frac{1}{25}} - 2\sqrt[3]{\frac{5}{64}}$

10. $7\sqrt[3]{\frac{1}{49}} + \sqrt[3]{\frac{1}{16}} + \sqrt[3]{\frac{1}{2}} - 2\sqrt[3]{\frac{7}{8}}$

11. $\frac{2}{3}\sqrt[3]{135} + \frac{1}{2}\sqrt[3]{\frac{1}{32}} + \frac{7}{4}\sqrt[3]{\frac{1}{4}} - 20\sqrt[3]{\frac{1}{200}}$

12. $3\sqrt[3]{-24} - 4\sqrt[3]{-81} - \sqrt[3]{-375}$

13. $4\sqrt[3]{-320} - 10\sqrt[3]{-40} - 2\sqrt[3]{-54} + 3\sqrt[3]{-1024}$

14. $3\sqrt[3]{2a^3} - b\sqrt[3]{128} + (4b - 3a)\sqrt[3]{2}$

15. $a\sqrt[3]{250b} - \sqrt[3]{3ab^3} - 5\sqrt[3]{2a^3b} + 3b\sqrt[3]{3a}$

11.- Racionaliza

1. $\frac{1}{\sqrt{3}}$

3. $\frac{3}{4\sqrt{5}}$

5. $\frac{5}{\sqrt[3]{4a^2}}$

7. $\frac{3}{\sqrt[4]{9a}}$

9. $\frac{x}{\sqrt[4]{27x^2}}$

11. $\frac{5n^2}{3\sqrt{mn}}$

2. $\frac{5}{\sqrt{2}}$

4. $\frac{2a}{\sqrt{2ax}}$

6. $\frac{1}{\sqrt[3]{9x}}$

8. $\frac{6}{5\sqrt[3]{3x}}$

10. $\frac{1}{\sqrt[5]{8a^4}}$

12. $\frac{1}{5a\sqrt[4]{25x^3}}$

12.- Racionalizar el denominador:

1. $\frac{3 - \sqrt{2}}{1 + \sqrt{2}}$

4. $\frac{\sqrt{7} + 2\sqrt{5}}{\sqrt{7} - \sqrt{5}}$

7. $\frac{3\sqrt{2}}{7\sqrt{2} - 6\sqrt{3}}$

10. $\frac{\sqrt{7} + 3\sqrt{11}}{5\sqrt{7} + 4\sqrt{11}}$

13. $\frac{\sqrt{a} + \sqrt{x}}{2\sqrt{a} + \sqrt{x}}$

16. $\frac{\sqrt{x+2} + \sqrt{x-2}}{\sqrt{x+2} - \sqrt{x-2}}$

2. $\frac{5 + 2\sqrt{3}}{4 - \sqrt{3}}$

5. $\frac{\sqrt{2} - 3\sqrt{5}}{2\sqrt{2} + \sqrt{5}}$

8. $\frac{4\sqrt{3} - 3\sqrt{7}}{2\sqrt{3} + 3\sqrt{7}}$

11. $\frac{\sqrt{5} + \sqrt{2}}{7 + 2\sqrt{10}}$

14. $\frac{\sqrt{x} - \sqrt{x-1}}{\sqrt{x} + \sqrt{x-1}}$

17. $\frac{\sqrt{a+4} - \sqrt{a-4}}{\sqrt{a+4} + \sqrt{a-4}}$

3. $\frac{\sqrt{2} - \sqrt{5}}{\sqrt{2} + \sqrt{5}}$

6. $\frac{19}{5\sqrt{2} - 4\sqrt{3}}$

9. $\frac{5\sqrt{2} - 6\sqrt{3}}{4\sqrt{2} - 3\sqrt{3}}$

12. $\frac{9\sqrt{3} - 3\sqrt{2}}{6 - \sqrt{6}}$

15. $\frac{\sqrt{a} - \sqrt{a+1}}{\sqrt{a} + \sqrt{a+1}}$

18. $\frac{\sqrt{a+b} - \sqrt{a-b}}{\sqrt{a+b} + \sqrt{a-b}}$

13.- Problemas

a) Calcula el área y el perímetro de un triángulo equilátero de lado a cm.

b) Calcula el volumen, el área y la diagonal de un cubo de arista 2 cm. Expresa el resultado con radicales.

c) Calcula el perímetro de la figura sombreada expresando el resultado con radicales. ¿Cuánto vale la cuarta parte de ese perímetro?

d) Se tiene una mesa de superficie cilíndrica de 2 m de diámetro y 1 m de altura (ver figura). Se quiere cubrir con un mantel cuadrado de manera que roce ligeramente el piso. ¿Cuál debe ser la medida que corresponde al lado del mantel?

